

Clinical Operations Analytics Platform

Microsoft Fabric Architecture Document

Prepared for: Regional Health Network — 12-Hospital System

Document Owner: Enterprise Data & Analytics (EDA)

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Document Control

Version	Date	Author / Role	Summary of Change
0.1	Week 1	Data Platform Architect	Initial skeleton, source system inventory
0.5	Week 2	Data Platform Architect	Medallion design, semantic model, capacity sizing
0.9	Week 3	Architect + Privacy Officer	Security model, RLS/OLS/CLS, HIPAA control mapping
1.0	Week 4	Enterprise Data & Analytics	Full draft released for Architecture Review Board sign-off

Required Reviewers and Approvers

Role	Responsibility in this Document	Approval Required
Chief Medical Information Officer	Validates that KPI definitions match clinical intent	Yes
HIPAA Privacy Officer	Approves PHI handling, minimum-necessary design, audit controls	Yes
Information Security Officer	Approves identity, access, encryption, and logging design	Yes
Director, Enterprise Data & Analytics	Owens platform delivery, capacity budget, and operations	Yes
EHR Application Manager	Confirms extract method, refresh window, and source SLAs	Yes
Hospital Operations Leadership	Confirms report content and departmental access model	Advisory

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3. Security and Governance — principles, identity, workspace roles, PHI classification, RLS, OLS/CLS, SQL GRANT and DENY, HIPAA compliance, governance

4. Capacity and Performance — workload profile, SKU recommendation, cost management, autoscaling strategy, performance targets

5. DevOps and Deployment — environment model, branching, Git integration, deployment pipeline, deployment rules, CI/CD triggers, release gates, change management

6. Operational Procedures — capacity monitoring, incident response, scheduled maintenance, business continuity

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1. Executive Summary

1.1 Purpose

This document defines the target architecture for a clinical operations analytics platform built on Microsoft Fabric for a regional healthcare network of twelve hospitals. It specifies the data architecture, security and governance model, capacity plan, deployment process, and operational procedures required to move the network from fragmented, hospital-by-hospital reporting to a single governed analytics platform that satisfies HIPAA obligations for Protected Health Information (PHI).

The document is written to be actionable: every section states a specific recommendation with the reasoning behind it, so that engineering can build from it and the Architecture Review Board, Privacy Officer, and Information Security Officer can approve or challenge concrete decisions rather than generalities.

1.2 The Clinical Operations Problem

The network currently produces operational reporting through hospital-local extracts from the Electronic Health Record (EHR), maintained largely in spreadsheets by individual decision-support analysts. Three consequences follow. First, the same measure is calculated differently at different hospitals — average length of stay is computed on discharges at some sites and on census days at others — so network-level comparison is unreliable. Second, reporting lags the clinical event by seven to ten days, which is too slow to influence throughput decisions. Third, extracts containing identifiable patient data circulate as files with no enforceable access control, which is a direct HIPAA exposure.

The platform is designed to answer a defined set of clinical operations questions consistently across all twelve hospitals:

- Length of stay: What is the average and median length of stay by hospital, department, service line, and discharging provider, and how does it trend against the case-mix-adjusted target?
- Readmissions: Which departments and service lines exceed the network benchmark for 30-day unplanned readmission, and which index conditions drive that variance?
- Encounter volume and demand: What is encounter volume by department, day of week, and hour of arrival, and where does arriving demand exceed staffed capacity?
- Emergency throughput: What is door-to-provider time, ED length of stay, boarding time for admitted patients, and left-without-being-seen rate by site?
- Capacity and flow: What is current bed occupancy and pending-discharge volume, and how does discharge timing distribute across the day?
- Disposition and continuity: What is the discharge disposition mix, observation-to-inpatient conversion rate, and rate of discharge against medical advice?

These questions define the scope of the gold layer and the certified semantic model. Any measure not traceable to one of them is out of scope for release 1.

1.3 Solution Approach

The platform is built entirely on Microsoft Fabric, using OneLake as the single storage layer and a Medallion (bronze / silver / gold) architecture to separate raw capture from clinical standardization and from analytics-ready aggregation. Patient encounter data is ingested daily from the EHR by Data Factory pipelines invoking watermark-driven Copy activity; a parallel Eventstream carries HL7 Admit-Discharge-Transfer (ADT) messages into an Eventhouse for near-real-time census and bed-occupancy visibility. PySpark notebooks perform cleansing, terminology mapping, and deduplication in silver, and build conformed dimensional and aggregate tables in gold.

A single certified semantic model sits on gold in Direct Lake mode and carries the network-approved definitions of every clinical KPI. Security is layered: Microsoft Entra ID groups drive Fabric workspace roles; row-level security (RLS) restricts each user to their own hospital, department, or provider panel; object-level and column-level security (OLS/CLS) remove PHI columns from any role that does not need identifiable data; and explicit SQL GRANT and DENY statements govern direct query access through the Lakehouse SQL analytics endpoint. All content is version-controlled in Git and promoted through Development, Test, and Production workspaces by Fabric deployment pipelines with environment-specific deployment rules, triggered on pull request completion.

1.4 Key Benefits

Four benefits justify the investment and should be the basis for measuring the platform after go-live.

1. Faster access to clinical data. Encounter data is available to department leadership by 06:00 the morning after the clinical day, replacing a seven-to-ten-day reporting lag, and ED census and bed occupancy refresh within five minutes of the underlying ADT event. Operational decisions about staffing and throughput can be made on the current day rather than retrospectively.
2. Consistent metrics across all twelve hospitals. One certified semantic model holds one definition of average length of stay, readmission rate, and encounter volume. Site-level comparison, service-line review, and board reporting all draw from the same measures, which removes the recurring reconciliation argument that currently consumes decision-support capacity.
3. Demonstrable HIPAA compliance. Minimum-necessary access is enforced technically rather than by policy alone: RLS confines users to their own department or panel, OLS and CLS strip identifiers from analyst-facing roles, and every access to identifiable data is logged and retained. The network can produce evidence of who accessed which patient data, when, and under what authorization — evidence that spreadsheet distribution cannot produce.
4. Lower total cost and faster delivery of new analytics. Consolidating hospital-local extracts onto a single Fabric capacity removes duplicated storage and duplicated engineering effort. At the F64 capacity recommended in Section 4, report consumers do not require individual Power BI Pro licenses, which materially reduces the cost of extending access from decision-support staff to frontline clinical leadership.

1.5 Assumptions

#	Assumption	Impact if Invalid
A1	The EHR vendor provides a supported clinical data warehouse or reporting database with a reliable last-modified timestamp for incremental extraction.	Ingestion shifts to full-table extract or vendor API, increasing refresh duration and CU consumption.
A2	The network operates a single Microsoft Entra ID tenant covering all twelve hospitals.	Cross-tenant or B2B identity design required; RLS bridge table and group strategy must be reworked.
A3	An enterprise master patient index (EMPI) supplies a stable enterprise patient identifier across sites.	Cross-site readmission measurement becomes unreliable; probabilistic matching would be required in silver.
A4	A Business Associate Agreement and the HIPAA-eligible services configuration are in place with Microsoft for Azure and Fabric.	Platform cannot legally hold PHI; project stops until executed.
A5	Capacity is deployed in a US Azure region approved by the network for PHI, with data residency confirmed.	Region change required; OneLake and capacity must be redeployed.
A6	Clinical department and provider hierarchies are maintained in a source of record that can be extracted daily.	RLS scope data becomes stale, producing both over- and under-permissioned users.

1.6 Out of Scope for Release 1

- Clinical quality registry submission, regulatory quality measure certification (for example eCQM abstraction), and payer claims analytics.
- Predictive or machine-learning models, including readmission risk scoring. The gold layer is designed to support these later, but no model is deployed in release 1.
- Revenue cycle, supply chain, and human resources subject areas.
- Write-back to the EHR. The platform is strictly read-only with respect to clinical source systems.
- Patient-facing or externally shared reporting. All distribution is internal to the network tenant.

2. Architecture Overview

2.1 Logical Architecture

The diagram below shows the end-to-end architecture: source systems on the left, the Fabric platform with its Medallion layers and semantic model in the centre, and consumption on the right. A written walkthrough follows in Section 2.3 for readers using assistive technology or a text-only copy of this document.

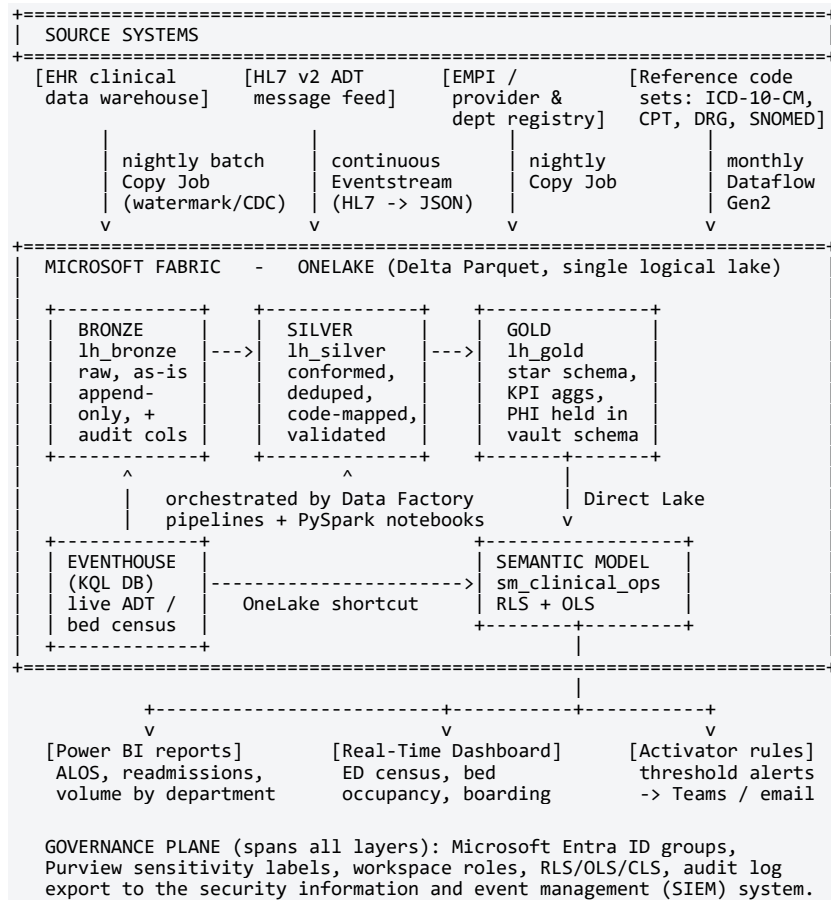


Figure 1 — Logical architecture, source systems through consumption.

2.2 Component Inventory

Component	Fabric Item Type	Purpose	Owner
lh_bronze	Lakehouse	Immutable landing zone for raw EHR encounter extracts, partitioned by load date	Data Engineering
lh_silver	Lakehouse	Cleansed, deduplicated, terminology-mapped clinical entities at encounter grain	Data Engineering
lh_gold	Lakehouse	Conformed star schema and pre-aggregated clinical KPI tables	Data Engineering

Component	Fabric Item Type	Purpose	Owner
pl_ehr_daily_load	Data Pipeline	Nightly orchestration: extract, validate, transform, publish, notify	Data Engineering
cj_ehr_encounters	Copy Job	Incremental extract of encounter and related detail tables from the EHR	Data Engineering
nb_silver_*, nb_gold_*	Notebook (PySpark)	Standardization, deduplication, dimensional build, KPI aggregation	Data Engineering
df_reference_codes	Dataflow Gen2	Analyst-maintained reference and mapping tables (low volume, monthly)	Clinical Informatics
es_adt_stream	Eventstream	Continuous ingestion and parsing of HL7 v2 ADT messages	Integration Team
eh_clinical_rt	Eventhouse (KQL)	Time-series store for live census, bed status, and boarding events	Data Engineering
sm_clinical_ops	Semantic Model	Certified Direct Lake model carrying all approved KPI definitions, RLS and OLS	Analytics Engineering
rpt_* (report set)	Power BI Report	Executive, hospital, department, and ED throughput reports	Analytics Engineering
rtd_capacity_now	Real-Time Dashboard	Live census and bed occupancy for house supervisors	Analytics Engineering
act_ops_alerts	Activator	Threshold alerting on census, boarding time, and data freshness	Operations

2.3 Data Flow: Sources to Reports

The following walkthrough describes Figure 1 in narrative form.

1. **Extraction.** At 01:00 local time, pipeline pl_ehr_daily_load starts. A Copy Job reads changed rows from the EHR clinical data warehouse using a high-water mark on the source last-modified timestamp, stored in a control table in lh_bronze. Each of the twelve hospitals is a partition of the same logical extract; the pipeline uses a ForEach over the hospital list with a bounded concurrency of four to avoid saturating the source system during its own maintenance window.
2. **Bronze landing.** Rows land in lh_bronze as Delta tables, appended without modification and stamped with technical audit columns: source system, extract batch identifier, source file or query fingerprint, and ingestion timestamp. Bronze is never edited or corrected in place. If a source correction arrives, it lands as a new row, and silver resolves precedence. This preserves an accurate record of what the EHR asserted at each point in time, which matters both for clinical defensibility and for HIPAA audit.
3. **Silver standardization.** PySpark notebooks read new bronze partitions and write conformed tables to lh_silver. This layer performs date and time normalization to a single time zone convention, terminology mapping to standard code sets, deduplication to one row per encounter using the enterprise patient identifier and source encounter identifier, referential validation against

dimension sources, and rejection of records failing mandatory-field or plausibility checks into a quarantine table for steward review.

4. **Gold modelling.** A second notebook set builds the conformed star schema in `lh_gold` — dimensions for patient, provider, department, hospital, date, diagnosis, and disposition, and fact tables at encounter grain plus daily census grain — and then materializes the pre-aggregated KPI tables that back the highest-traffic report pages. Identifiable patient attributes are written to a separate restricted schema (referred to in this document as the PHI vault) rather than being mixed into the general dimension tables.
5. **Semantic model refresh.** The certified semantic model `sm_clinical_ops` sits directly on gold Delta tables in Direct Lake mode, so no data copy or scheduled import refresh is required; the pipeline issues a framing operation at the end of the run so the model picks up the new Delta version. RLS roles and OLS rules are defined in the model and evaluated per user at query time.
6. **Consumption.** Power BI reports connect live to the certified model. A separate real-time dashboard queries the Eventhouse for live census. Activator rules watch both paths and raise alerts to Teams and email when thresholds are crossed — including a data-freshness rule that fires if gold has not been updated by 06:00.
7. **Notification and close-out.** The pipeline writes run statistics to an operational log table, posts a completion summary to the data engineering Teams channel, and, on failure, raises an incident through the process defined in Section 6.2.

2.4 Ingestion Approach and Rationale

Fabric offers three principal ingestion mechanisms. They are not interchangeable, and the choice materially affects refresh duration, capacity consumption, and maintainability. The table below records the evaluation.

Mechanism	Strengths	Limitations for this Workload	Verdict
Data Pipeline (Data Factory)	Full orchestration: parameterization, ForEach over hospitals, conditional logic, retries, failure handling, notebook and Copy activity invocation	Not itself a transformation engine; needs notebooks for real cleansing work	Selected as the orchestrator for the entire nightly load
Copy Job	Purpose-built for repeatable incremental copy; native change-data-capture and watermark handling; low authoring overhead; efficient on large tables	Copy only — no transformation; limited to supported source connectors	Selected for the bulk EHR extract and EMPI/registry extract, invoked by the pipeline
Dataflow Gen2	Low-code Power Query authoring accessible to clinical informatics analysts; good for small, business-curated data	Higher CU cost per row at volume; harder to unit test and code-review; performance degrades on large clinical fact loads	Restricted to small reference and mapping tables owned by informatics

Mechanism	Strengths	Limitations for this Workload	Verdict
Eventstream	Continuous ingestion of HL7 v2 ADT with sub-minute latency into Eventhouse	Not appropriate for bulk historical clinical detail	Selected for the real-time ADT path only
Notebook (PySpark)	Full expressive power for deduplication, terminology mapping, and dimensional builds; testable, reviewable, source-controlled	Requires engineering skill set; not analyst-authored	Selected for all bronze-to-silver and silver-to-gold transformation

Recommendation: a Data Factory pipeline orchestrates the nightly run; Copy Job performs the incremental EHR extract into bronze; PySpark notebooks perform all transformation; Dataflow Gen2 is permitted only for low-volume reference data maintained by clinical informatics; Eventstream serves the real-time ADT path. This split gives engineering-grade control over the clinical data path while still allowing informatics analysts to maintain mapping tables without filing engineering tickets.

Incremental strategy: the extract is watermark-based on the source last-modified timestamp, with a deliberate 48-hour lookback window on each run. Clinical source systems routinely backdate corrections — a coding update or a late discharge disposition entered two days after the fact — and a naive high-water mark would silently miss them. Silver applies last-write-wins precedence within the lookback window, so restated records correct themselves without a full reload.

2.5 Medallion Layer Design for Clinical Data

Bronze — raw clinical capture

Bronze is an append-only, schema-on-read reproduction of what the source supplied, with no business logic applied. Tables are partitioned by ingestion date. Retention is seven years to align with the network medical record retention policy, with data older than thirteen months moved to cool storage. Bronze is accessible only to the data engineering service principal and to break-glass administrators; no analyst and no report ever reads bronze.

Bronze Table	Grain	Approximate Daily Rows	Notes
br_encounter	One row per encounter version	9,000	Inpatient, observation, ED, and outpatient encounters
br_encounter_diagnosis	Encounter × diagnosis	95,000	Includes present-on-admission indicator
br_encounter_procedure	Encounter × procedure	38,000	CPT and ICD-10-PCS as supplied
br_adt_event	ADT event	52,000	Admission, transfer, discharge, bed assignment
br_patient	Patient version	11,000	Contains raw identifiers; most restricted table on the platform

Bronze Table	Grain	Approximate Daily Rows	Notes
br_provider	Provider version	400	Includes department and service line attribution
br_department	Department version	60	Hospital, department, service line hierarchy

Silver — conformed clinical data

Silver applies the standardization rules that make twelve hospitals comparable. The specific transformations are:

- Date and time normalization. All timestamps are stored in Coordinated Universal Time with a companion local-time column and the site time zone, so overnight boundaries and daylight-saving transitions do not distort length-of-stay or arrival-hour analysis.
- Terminology mapping. Site-local codes for department, encounter class, discharge disposition, and admission source are mapped to the network standard value set. Unmapped codes are routed to a quarantine table and reported daily to clinical informatics rather than silently defaulted, because a silent default here corrupts the very comparisons the platform exists to make.
- Deduplication. One row per encounter is retained, keyed on enterprise patient identifier plus source encounter identifier, resolving to the most recent source version within the lookback window.
- Patient identity resolution. The EMPI enterprise identifier is attached to every encounter so that a patient treated at Hospital A and readmitted to Hospital B is recognized as one person — a prerequisite for network-level readmission measurement.
- Clinical validity checks. Discharge before admission, negative length of stay, implausible ages, and encounters referencing unknown departments are rejected to quarantine with a reason code.
- PHI segregation begins here. Direct identifiers are written to a restricted silver schema, and downstream tables carry surrogate keys, so the default path for any consumer is de-identified.

Gold — clinical KPI layer

Gold is a Kimball-style star schema plus a small set of aggregate tables. The fact tables are:

Gold Table	Type	Grain	Supports
fact_encounter	Fact	One row per completed encounter	ALOS, volume, disposition mix, observation conversion
fact_daily_census	Periodic snapshot	Hospital × department × date	Occupancy, midnight census, staffed-bed utilization
fact_readmission	Accumulating snapshot	Index discharge with linked readmission	30-day unplanned readmission rate

Gold Table	Type	Grain	Supports
fact_ed_throughput	Fact	One row per ED encounter	Door-to-provider, ED LOS, boarding, LWBS
dim_patient	Dimension (SCD-2)	Patient version	De-identified demographics; identifiers held in vault
dim_provider	Dimension (SCD-2)	Provider version	Provider, department, service line attribution
dim_department	Dimension (SCD-2)	Department version	Hospital, department, service line hierarchy; drives RLS
dim_hospital	Dimension	Hospital	Site, region, bed size, teaching status
dim_date	Dimension	Calendar date	Fiscal calendar, day of week, holiday flag
agg_kpi_dept_day	Aggregate	Hospital × department × date	Pre-aggregated headline KPIs for fast page loads

Gold tables are written with V-Order optimization enabled and OPTIMIZE with compaction run after each load, so that Direct Lake queries read a small number of well-sized row groups rather than many small files. Partitioning is by discharge date for encounter facts and by census date for the snapshot fact.

2.6 Semantic Model Design

One certified semantic model, `sm_clinical_ops`, serves all reports. It is deliberately singular: multiple models would reintroduce exactly the metric divergence the platform exists to eliminate. The model is configured in Direct Lake mode over gold, with a fallback to DirectQuery disabled so that a silent fallback cannot degrade performance without anyone noticing.

Model structure follows a strict star: fact tables related to dimensions on surrogate keys, single-direction filters from dimension to fact, no bidirectional relationships except through the security bridge described in Section 3.5, and no calculated columns on fact tables. Representative measure definitions:

```

Discharges =
    CALCULATE(
        COUNTROWS( fact_encounter ),
        fact_encounter[EncounterStatus] = "Discharged",
        fact_encounter[EncounterClass] IN { "Inpatient", "Observation" }
    )

Patient Days =
    SUM( fact_encounter[LengthOfStayDays] )

Average Length of Stay =
    DIVIDE( [Patient Days], [Discharges] )

Index Discharges =
    CALCULATE(
        COUNTROWS( fact_readmission ),
        fact_readmission[IsEligibleIndex] = TRUE()
    )

30-Day Unplanned Readmissions =
    CALCULATE(

```

```

COUNTROWS( fact_readmission ),
fact_readmission[IsEligibleIndex] = TRUE(),
fact_readmission[ReadmitWithin30Days] = TRUE(),
fact_readmission[IsPlannedReadmission] = FALSE()
)

30-Day Readmission Rate =
    DIVIDE( [30-Day Unplanned Readmissions], [Index Discharges] )

Encounter Volume =
    COUNTROWS( fact_encounter )

Occupancy Rate =
    DIVIDE(
        SUM( fact_daily_census[MidnightCensus] ),
        SUM( fact_daily_census[StaffedBeds] )
    )

-- Small-cell suppression: never display a rate derived from fewer than
-- 11 encounters, to prevent re-identification in thinly populated
-- department-by-month cells.
30-Day Readmission Rate (Safe) =
    IF( [Index Discharges] < 11, BLANK(), [30-Day Readmission Rate] )

```

The small-cell suppression pattern shown in the final measure is applied to every rate measure exposed at department, provider, or month grain. A readmission rate computed on four encounters is both clinically meaningless and a genuine re-identification risk in a small specialty department, and suppression at the measure layer is more reliable than asking report authors to remember.

2.7 Real-Time Component

Length of stay and readmission are retrospective measures and are well served by the nightly batch. Bed occupancy and ED boarding are not — a house supervisor making a bed assignment at 14:00 cannot use last night data. A real-time path is therefore in scope for release 1, but is deliberately narrow.

- **Ingestion:** the existing integration engine forwards a copy of HL7 v2 ADT messages (A01 admit, A02 transfer, A03 discharge, A08 update, A11/A13 cancellations) to an Eventstream, which parses them to JSON.
- **Storage:** events land in Eventhouse (KQL database) eh_clinical_rt with a 90-day hot retention, sufficient for operational use; the authoritative long-term record remains the batch path.
- **Consumption:** a Real-Time Dashboard shows live census by unit, admitted-patients-awaiting-bed count, and boarding duration. Activator rules raise alerts to the house supervisor Teams channel when boarding time exceeds four hours or a unit exceeds ninety-five percent occupancy.
- **Reconciliation:** the real-time path is explicitly labelled as operational and non-authoritative. Every real-time dashboard carries a footer stating that figures are provisional and that the certified batch model is the system of record for reported metrics. A daily reconciliation job compares real-time census against the batch snapshot and flags variance above two percent for investigation.

The same PHI rules apply to the real-time path. The Eventhouse holds the enterprise patient identifier but not patient name or date of birth; the dashboard displays bed and unit-level counts rather than patient-level detail.

2.8 Workspace Topology

Workspace	Capacity	Contents	Git Branch
ws-clinical-dev	F8 (dev)	Lakehouses, pipelines, notebooks, model, reports — synthetic and de-identified data only	feature/* and main
ws-clinical-test	F16 (test)	Full item set; masked production-like data volume for validation	main (release candidate)
ws-clinical-prod	F64 (prod)	Full item set; live PHI	main (tagged release)
ws-clinical-rt-prod	F64 (prod)	Eventstream, Eventhouse, real-time dashboard	main
ws-clinical-sandbox	F8 (dev)	Analyst exploration on de-identified extracts; not promoted	not source-controlled

Separating the real-time items into their own production workspace lets the streaming path be redeployed or restarted without touching the batch platform, and lets its access list stay smaller.

3. Security and Governance

3.1 Security Design Principles

The security model is built on the assumption that a single control will eventually fail, and that no single control should be the only thing standing between a user and identifiable patient data. Five layers apply in sequence, and a user must pass all of them to see a value.

Layer	Control	What It Restricts
1	Microsoft Entra ID group membership	Whether the person can authenticate to the platform at all
2	Fabric workspace role	Which items the person can see, edit, or publish
3	Row-level security (RLS)	Which patient encounter rows the person can retrieve
4	Object- and column-level security (OLS/CLS)	Which PHI attributes exist for that person at all
5	SQL GRANT / DENY on the SQL analytics endpoint	Whether direct T-SQL query bypasses the semantic model

Two rules govern the whole design. First, deny by default: a new user receives no access until assigned to a group, and a new PHI column is inaccessible until explicitly granted. Second, no standing broad access: nobody holds network-wide identifiable data access as a routine entitlement; that access is time-bound, justified, and logged.

3.2 Identity and Group Structure

All access is granted to Microsoft Entra ID security groups, never to individual users. Groups are populated from the HR and provider credentialing systems where possible, so that a transfer or termination propagates automatically. Individual assignment is permitted only for break-glass administrator accounts, which are separately monitored.

Entra ID Group	Population	Intended Platform Access
SG-Fabric-Clinical-Admins	Platform architects (3 named)	Workspace Admin, production
SG-Fabric-Clinical-Engineers	Data engineering team (8)	Contributor in dev/test, none in prod
SG-Fabric-Clinical-Analysts	Decision support analysts (26)	Viewer, de-identified network scope
SG-Fabric-Clinical-DeptHeads	Department heads and managers (140)	Viewer, RLS-scoped to own department
SG-Fabric-Clinical-Providers	Attending physicians (620)	Viewer, RLS-scoped to own panel
SG-Fabric-Clinical-Executives	Hospital and network executives (32)	Viewer, network scope, de-identified
SG-Fabric-Clinical-Compliance	Privacy and compliance officers (6)	Viewer with audit access, PHI-eligible
SG-Fabric-Clinical-SvcPrincipals	Automation identities (4)	Contributor via service principal, prod pipelines
SG-Fabric-Clinical-BreakGlass	Emergency access accounts (2)	Admin, PIM-activated, 4-hour maximum

Privileged roles — workspace Admin and any PHI-eligible role — are assigned through Microsoft Entra Privileged Identity Management with activation requiring justification, approval, and multi-factor authentication, and a maximum activation duration of four hours. Standing membership in these roles is prohibited.

3.3 Workspace Roles and Assignments

Fabric provides four workspace roles. Their practical meaning for this platform is:

- Admin — full control including workspace deletion, access management, and capacity assignment. Restricted to three named platform architects, activated through PIM.
- Member — can add other users up to Member level and manage all content. Deliberately unused in production, because it allows access re-granting outside the change process.
- Contributor — can create and modify items but cannot manage access. This is the working role for engineers in development and test.
- Viewer — can read reports and query the semantic model, subject to RLS and OLS. This is the role for every clinical consumer.

Group / Persona	Dev	Test	Prod	Rationale
Platform architects	Admin	Admin	Admin (PIM)	Own the workspace; production activation is time-bound and approved
Data engineering team	Contributor	Contributor	None	Build in dev, validate in test; production changes arrive only through the deployment pipeline, never by hand
Deployment service principal	Contributor	Contributor	Contributor	The only identity that writes to production, and it does so from a reviewed, versioned artifact
Analytics engineers (model/report)	Contributor	Contributor	None	Same promotion discipline as engineering
Clinical analysts (decision support)	Viewer	Viewer	Viewer	Read certified content; RLS network-wide, OLS removes identifiers
Department heads	None	None	Viewer	Read reports scoped by RLS to their own department
Attending providers	None	None	Viewer	Read reports scoped by RLS to their own patient panel
Executives	None	None	Viewer	Network-wide aggregate view, no identifiers
Compliance / privacy officers	None	Viewer	Viewer	Read-only with audit-scope role; PHI-eligible for investigations
Clinical informatics (reference data)	Contributor	None	None	Maintain mapping Dataflows in dev only; changes promote like code

Two decisions in this table warrant emphasis for the review board. First, no human holds Contributor in production. Every production change flows through the deployment pipeline described in Section 5, which means every production change has a pull request, a reviewer, and an audit trail. Second, compliance officers hold Viewer rather than Admin. Auditors need to see data and access history; they do not need the ability to change the platform, and granting them Admin would weaken the very separation of duties they exist to verify.

3.3.1 Item-level and OneLake permissions

Workspace roles are supplemented by item-level sharing so that Viewer in the production workspace does not become a back door to raw data. Specifically: lh_bronze and lh_silver have OneLake data access removed for the Viewer role, so a workspace Viewer cannot open the bronze Lakehouse or query its SQL endpoint; lh_gold grants read only through the semantic model, not through direct file access; and the semantic model is shared with the Build permission required for report consumption but not the write permission that would allow a user to alter measure definitions.

3.4 PHI Classification

Every column in gold is classified before release. Classification drives the OLS, CLS, and masking rules that follow, and is recorded as a Microsoft Purview sensitivity label on the item and as a column-level tag in the data catalog.

Column	Classification	Treatment in Platform
PatientName	Direct identifier (PHI)	Vault schema only; OLS-hidden from all roles except Compliance
SocialSecurityNumber	Direct identifier (PHI)	Not loaded to gold at all; remains in bronze restricted schema
MedicalRecordNumber	Direct identifier (PHI)	Vault schema; masked for all roles except Compliance
DateOfBirth	Direct identifier (PHI)	Replaced in gold by AgeAtEncounter and AgeBand; ages over 89 collapsed to a single 90+ band
Address / ZIP code	Direct identifier (PHI)	Truncated to three-digit ZIP; low-population ZIPs collapsed
PhoneNumber, Email	Direct identifier (PHI)	Not loaded to gold
EnterprisePatientID	Pseudonymous key	Surrogate key used for joins; never displayed in reports
ProviderNPI, ProviderName	Sensitive (not PHI)	Visible to Department Heads for own department; suppressed below small-cell threshold
DiagnosisCode, ProcedureCode	Clinical (PHI in combination)	Available to all clinical roles; safe in aggregate, suppressed at small cell counts
LengthOfStayDays, Disposition	Clinical (PHI in combination)	Available to all clinical roles

The design point behind this table is that the default analytic path is de-identified. Date of birth is not hidden in gold — it is not present in gold, having been converted to an age band during the silver-to-gold

build. A control that removes data is stronger than a control that hides it, because it cannot be misconfigured back on.

3.5 Row-Level Security Design

RLS is dynamic and data-driven. Rather than creating one static role per department — which would mean maintaining 60 roles across 12 hospitals — a single security bridge table maps each user principal name to the scopes they are entitled to, and a small number of DAX roles read that table at query time.

3.5.1 The security scope table

Table `sec_user_scope` is built nightly in gold from the HR system, the provider credentialing system, and the department registry. It is the single source of truth for who sees what.

Column	Type	Description
UserPrincipalName	string	Entra ID UPN, lower-cased, the join key to USERPRINCIPALNAME()
ScopeType	string	One of Department, Hospital, Panel, Network
HospitalKey	int	Surrogate key of the hospital in scope; -1 for network scope
DepartmentKey	int	Surrogate key of the department in scope; -1 for hospital or network scope
ProviderKey	int	Surrogate key of the provider whose panel is in scope; -1 if not panel-scoped
EffectiveFrom / EffectiveTo	date	Supports scheduled onboarding and automatic expiry on transfer or termination
GrantedBy / GrantReason	string	Retained for HIPAA audit: who authorized this access and why

The table is filtered to rows where the current date falls between `EffectiveFrom` and `EffectiveTo`, so a department head who moves to another service loses the old scope automatically on the effective date rather than waiting for a manual review.

3.5.2 RLS role definitions

RLS Role	Assigned Group	Row Scope	Identifiers Visible
RLS_DepartmentHead	SG-Fabric-Clinical-DeptHeads	Encounters in the user own department(s) only	No
RLS_Provider	SG-Fabric-Clinical-Providers	Encounters where the user is attending or admitting provider	MRN only
RLS_HospitalLeader	SG-Fabric-Clinical-Executives (site)	All encounters at the user own hospital	No
RLS_NetworkAnalyst	SG-Fabric-Clinical-Analysts	All encounters, all sites	No
RLS_Executive	SG-Fabric-Clinical-Executives (network)	All encounters, all sites, aggregate reports only	No
RLS_Compliance	SG-Fabric-Clinical-Compliance	All encounters, all sites	Yes (logged)

3.5.3 DAX filter expressions

Filters are applied to the dimension tables, and propagate to facts through the model relationships. Applying the predicate to dim_department rather than to fact_encounter is deliberate: the dimension is small, the filter evaluates once, and every fact related to that dimension inherits the restriction, so a new fact table added later is protected by default.

```
-- Role: RLS_DepartmentHead
-- Applied to table: dim_department
VAR CurrentUser = LOWER( USERPRINCIPALNAME() )
RETURN
    dim_department[DepartmentKey] IN
        SELECTCOLUMNS(
            FILTER(
                sec_user_scope,
                LOWER( sec_user_scope[UserPrincipalName] ) = CurrentUser
                && sec_user_scope[ScopeType] = "Department"
                && sec_user_scope[EffectiveFrom] <= TODAY()
                && sec_user_scope[EffectiveTo] >= TODAY()
            ),
            "DeptKey", sec_user_scope[DepartmentKey]
        )

-- Role: RLS_HospitalLeader
-- Applied to table: dim_hospital
VAR CurrentUser = LOWER( USERPRINCIPALNAME() )
RETURN
    dim_hospital[HospitalKey] IN
        SELECTCOLUMNS(
            FILTER(
                sec_user_scope,
                LOWER( sec_user_scope[UserPrincipalName] ) = CurrentUser
                && sec_user_scope[ScopeType] = "Hospital"
                && sec_user_scope[EffectiveFrom] <= TODAY()
                && sec_user_scope[EffectiveTo] >= TODAY()
            ),
            "HospKey", sec_user_scope[HospitalKey]
        )

-- Role: RLS_Provider
-- Applied to table: dim_provider
VAR CurrentUser = LOWER( USERPRINCIPALNAME() )
RETURN
    dim_provider[ProviderKey] IN
        SELECTCOLUMNS(
            FILTER(
                sec_user_scope,
                LOWER( sec_user_scope[UserPrincipalName] ) = CurrentUser
                && sec_user_scope[ScopeType] = "Panel"
                && sec_user_scope[EffectiveFrom] <= TODAY()
                && sec_user_scope[EffectiveTo] >= TODAY()
            ),
            "ProvKey", sec_user_scope[ProviderKey]
        )

-- Roles: RLS_NetworkAnalyst, RLS_Executive, RLS_Compliance
-- Applied to table: sec_user_scope (guard filter only)
-- No filter is placed on dim_department, dim_hospital or dim_provider,
-- so all rows are visible. The guard below still requires the user to
-- hold an active Network scope row, so a lapsed or revoked entitlement
-- returns an empty result set rather than the whole network.
VAR CurrentUser = LOWER( USERPRINCIPALNAME() )
RETURN
    CALCULATE(
        COUNTROWS( sec_user_scope ),
        LOWER( sec_user_scope[UserPrincipalName] ) = CurrentUser,
        sec_user_scope[ScopeType] = "Network",
        sec_user_scope[EffectiveFrom] <= TODAY(),
        sec_user_scope[EffectiveTo] >= TODAY()
    ) > 0
```

Worked example. A Cardiology department head at Hospital 04 holds one row in `sec_user_scope` with `ScopeType` Department and the `DepartmentKey` for Cardiology at Hospital 04. When she opens the length-of-stay report, the `RLS_DepartmentHead` filter reduces `dim_department` to that one row; the relationship to `fact_encounter` propagates the restriction; and every visual on the page — including the network comparison tile — returns only Cardiology encounters at her site. If she is also given oversight of Cardiac Surgery, a second row is added to `sec_user_scope` and both departments appear the following day with no model change and no redeployment.

3.5.4 RLS implementation rules

- Users must never be workspace Admin, Member, or Contributor in the production workspace, because those roles bypass RLS entirely. This is the single most common cause of RLS failure in practice, and the role table in Section 3.3 is written to prevent it.
- The `sec_user_scope` table itself is hidden from report view and protected by its own filter so that a user cannot read the entitlements of others.
- A user with no matching scope row sees no data, not all data. This fail-closed behaviour is verified by an automated test in the deployment pipeline.
- Every RLS role is validated after each deployment using the View as role function plus a scripted DAX query executed as a set of test principals, with expected row counts asserted. Section 5.7 defines this as a release gate.

3.6 Object-Level and Column-Level Security for PHI

RLS controls which rows a user sees. It does nothing about which columns. A department head correctly restricted to Cardiology encounters would still see patient name on every one of those rows unless a second control is applied. OLS in the semantic model and CLS at the SQL endpoint provide that control.

3.6.1 Object-level security in the semantic model

OLS is defined per role and applied to the PHI vault table and its columns. When a column is OLS-restricted, it does not merely return blank — it is absent from the metadata for that user, so it cannot be selected, referenced in a measure, or discovered through field-list exploration. Any visual that references a restricted column returns an error for that role rather than silently exposing data, so OLS-restricted fields are kept off shared report pages and placed on role-specific pages.

Role	PatientName	MedicalRecordNumber	DateOfBirth	AgeBand
RLS_Compliance	Visible	Visible	Visible	Visible
RLS_Provider	Hidden	Visible (own panel)	Not in model	Visible
RLS_DepartmentHead	Hidden	Hidden	Not in model	Visible
RLS_HospitalLeader	Hidden	Hidden	Not in model	Visible
RLS_NetworkAnalyst	Hidden	Hidden	Not in model	Visible
RLS_Executive	Hidden	Hidden	Not in model	Visible

The provider exception is intentional and clinically necessary: a physician reviewing her own panel needs the medical record number to act on a finding, and she already has treatment-relationship access to those same patients in the EHR. Extending it no further than her own panel is exactly the minimum-necessary standard applied correctly.

OLS is expressed in the model metadata. The TMSL fragment below shows the pattern applied to the department head role; it is generated and applied by script during deployment rather than clicked in a user interface, so it is version-controlled and reviewable.

```
{
  "roles": [
    {
      "name": "RLS_DepartmentHead",
      "modelPermission": "read",
      "tablePermissions": [
        {
          "name": "dim_patient_vault",
          "metadataPermission": "none"
        },
        {
          "name": "dim_patient",
          "columnPermissions": [
            { "name": "PatientName", "metadataPermission": "none" },
            { "name": "MedicalRecordNumber", "metadataPermission": "none" }
          ]
        }
      ]
    }
  ]
}
```

3.6.2 Column-level security and masking at the SQL endpoint

OLS protects the semantic model. It does not protect a user who connects a SQL client directly to the Lakehouse SQL analytics endpoint. CLS and dynamic data masking close that path.

```
-- Column-level security: grant only the columns a role needs.
-- Granting at column level implicitly denies the ungranted columns.

GRANT SELECT ON gold.dim_patient
  ( PatientKey, AgeAtEncounter, AgeBand, SexAtBirth,
    ZipCode3, PrimaryLanguage )
  TO [SG-Fabric-Clinical-DeptHeads];

GRANT SELECT ON gold.dim_patient
  ( PatientKey, AgeAtEncounter, AgeBand, SexAtBirth,
    ZipCode3, PrimaryLanguage, MedicalRecordNumber )
  TO [SG-Fabric-Clinical-Providers];

-- Compliance receives the full column set, including direct identifiers.
GRANT SELECT ON gold.dim_patient_vault
  TO [SG-Fabric-Clinical-Compliance];

-- Dynamic data masking as a second line of defence, so that a
-- misconfigured grant still does not surface a usable identifier.
ALTER TABLE gold.dim_patient_vault
  ALTER COLUMN MedicalRecordNumber
  ADD MASKED WITH (FUNCTION = 'partial(0,"XXXXXX",4)');

ALTER TABLE gold.dim_patient_vault
  ALTER COLUMN PatientName
  ADD MASKED WITH (FUNCTION = 'default()');

GRANT UNMASK TO [SG-Fabric-Clinical-Compliance];
```

Row-level security is also enforced at the SQL endpoint, so that the department restriction holds for direct query users exactly as it does in the semantic model. Without this, a department head with SQL access could simply query around the model.

```
CREATE SCHEMA sec;
GO

CREATE FUNCTION sec.fn_department_predicate(@DepartmentKey INT)
    RETURNS TABLE
    WITH SCHEMABINDING
AS
RETURN
    SELECT 1 AS fn_result
    WHERE EXISTS (
        SELECT 1
        FROM gold.sec_user_scope AS s
        WHERE s.UserPrincipalName = LOWER(USER_NAME())
              AND s.ScopeType = 'Department'
              AND s.DepartmentKey = @DepartmentKey
              AND CAST(GETUTCDATE() AS DATE)
                BETWEEN s.EffectiveFrom AND s.EffectiveTo
        )
    OR IS_ROLEMEMBER('db_network_wide_read') = 1;
GO

CREATE SECURITY POLICY sec.encounter_department_filter
    ADD FILTER PREDICATE sec.fn_department_predicate(DepartmentKey)
    ON gold.fact_encounter,
    ADD FILTER PREDICATE sec.fn_department_predicate(DepartmentKey)
    ON gold.fact_daily_census,
    ADD FILTER PREDICATE sec.fn_department_predicate(DepartmentKey)
    ON gold.fact_ed_throughput
WITH (STATE = ON);
GO
```

3.7 SQL GRANT and DENY Statements

The statements below govern the Lakehouse SQL analytics endpoint and Warehouse. They complement — and must not contradict — the workspace roles in Section 3.3. Where the two overlap, the more restrictive outcome must prevail, and Section 3.7.2 defines the audit that verifies this.

3.7.1 Grant and deny script

```
-----
-- 1. Database roles: grant to roles, never to individual users.
-----
CREATE ROLE db_clinical_read_deid;      -- de-identified analytics
CREATE ROLE db_clinical_read_phi;      -- identifiable, restricted
CREATE ROLE db_network_wide_read;      -- bypasses department RLS
CREATE ROLE db_etl_service;            -- automation identities only
CREATE ROLE db_audit_read;             -- compliance, logs only
GO

ALTER ROLE db_clinical_read_deid ADD MEMBER [SG-Fabric-Clinical-Analysts];
ALTER ROLE db_clinical_read_deid ADD MEMBER [SG-Fabric-Clinical-DeptHeads];
ALTER ROLE db_network_wide_read ADD MEMBER [SG-Fabric-Clinical-Analysts];
ALTER ROLE db_clinical_read_phi ADD MEMBER [SG-Fabric-Clinical-Compliance];
ALTER ROLE db_audit_read ADD MEMBER [SG-Fabric-Clinical-Compliance];
ALTER ROLE db_etl_service ADD MEMBER [svc-fabric-clinical-etl];
GO

-----
-- 2. Baseline: no access to anything outside gold.
-----
DENY SELECT ON SCHEMA::bronze TO db_clinical_read_deid,
                                db_clinical_read_phi,
                                db_network_wide_read,
                                db_audit_read;

DENY SELECT ON SCHEMA::silver TO db_clinical_read_deid,
                                db_clinical_read_phi,
```

```

                                db_network_wide_read,
                                db_audit_read;
GO

-----
-- 3. De-identified analytics: read gold facts and dimensions,
--    but never the PHI vault.
-----
GRANT SELECT ON SCHEMA::gold          TO db_clinical_read_deid;
DENY  SELECT ON gold.dim_patient_vault TO db_clinical_read_deid;
DENY  SELECT ON gold.sec_user_scope    TO db_clinical_read_deid;

-- Belt and braces: deny the specific identifier columns even if a
-- future schema change surfaces them in a general gold table.
DENY SELECT ON gold.dim_patient ( PatientName, MedicalRecordNumber,
                                   DateOfBirth, SocialSecurityNumber )
    TO db_clinical_read_deid;
GO

-----
-- 4. PHI-eligible access: compliance and privacy investigation only.
-----
GRANT SELECT ON SCHEMA::gold          TO db_clinical_read_phi;
GRANT SELECT ON gold.dim_patient_vault TO db_clinical_read_phi;
GRANT UNMASK                                TO db_clinical_read_phi;
DENY INSERT, UPDATE, DELETE ON SCHEMA::gold TO db_clinical_read_phi;
GO

-----
-- 5. ETL service principal: write to gold, read everything upstream,
--    but no rights to change security objects.
-----
GRANT SELECT                                ON SCHEMA::bronze TO db_etl_service;
GRANT SELECT, INSERT, UPDATE, DELETE ON SCHEMA::silver TO db_etl_service;
GRANT SELECT, INSERT, UPDATE, DELETE ON SCHEMA::gold TO db_etl_service;
DENY ALTER, CONTROL                                ON SCHEMA::sec TO db_etl_service;
DENY ALTER ANY SECURITY POLICY                    TO db_etl_service;
GO

-----
-- 6. Audit access: log tables only, no clinical data.
-----
GRANT SELECT ON SCHEMA::audit TO db_audit_read;
DENY SELECT ON SCHEMA::gold TO db_audit_read;
GO

-----
-- 7. Universal prohibitions.
-----
DENY ALTER ANY USER, ALTER ANY ROLE TO db_clinical_read_deid,
                                         db_clinical_read_phi,
                                         db_network_wide_read,
                                         db_audit_read;

DENY VIEW DEFINITION ON SCHEMA::sec TO db_clinical_read_deid,
                                         db_network_wide_read;
GO

```

Two conventions are worth stating explicitly. DENY beats GRANT in SQL Server permission resolution, which is why the pattern throughout is to grant broadly at schema level and then deny narrowly at the sensitive object. And permissions are granted to database roles rather than to Entra groups directly, so that changing what a persona can do is a single role edit rather than a hunt through scattered object grants.

3.7.2 Reconciling SQL permissions with workspace roles

Workspace roles and SQL grants are two independent permission systems over the same data, and they can disagree. A workspace Viewer with OneLake data access retained could read gold files directly

regardless of a SQL DENY; conversely a SQL GRANT is meaningless to a user with no workspace access at all. Three rules resolve this.

1. The workspace role establishes the ceiling. No SQL GRANT can widen access beyond what the workspace role permits, and OneLake data access is removed for Viewer on bronze and silver so that file-level reads cannot bypass the SQL layer.
2. SQL DENY establishes the floor. Where a user holds workspace access that would otherwise permit an object, an explicit DENY on that object removes it.
3. Overlaps are audited, not assumed. A monthly reconciliation job enumerates effective permissions from both systems for every Entra group and reports any principal whose effective SQL access exceeds its intended persona. The report goes to the Information Security Officer and the Privacy Officer. Any discrepancy is treated as a finding with a fourteen-day remediation deadline, and unresolved findings escalate to the Architecture Review Board.

```
-- Monthly effective-permission extract feeding the reconciliation report
SELECT  pr.name                AS principal_name,
        pr.type_desc           AS principal_type,
        pe.state_desc          AS grant_or_deny,
        pe.permission_name,
        SCHEMA_NAME(o.schema_id) AS schema_name,
        o.name                 AS object_name,
        c.name                 AS column_name
FROM    sys.database_permissions AS pe
JOIN    sys.database_principals AS pr ON pr.principal_id = pe.grantee_principal_id
LEFT JOIN sys.objects           AS o  ON o.object_id    = pe.major_id
LEFT JOIN sys.columns           AS c  ON c.object_id    = pe.major_id
        AND c.column_id        = pe.minor_id
WHERE   pr.name LIKE 'db_%' OR pr.name LIKE 'SG-Fabric-%'
ORDER BY principal_name, schema_name, object_name, column_name;
```

3.8 HIPAA Compliance Considerations

3.8.1 Minimum necessary

The HIPAA minimum necessary standard requires that access be limited to the least PHI needed to accomplish the purpose. The platform implements this through the layered controls already described, and the mapping is explicit so it can be evidenced during an audit: role-based workspace assignment limits who reaches the platform; RLS limits which patients each role can see; OLS and CLS limit which attributes exist for that role; and de-identification during the silver-to-gold build means most analytic work never touches identifiers at all. Access requests are reviewed against a documented justification, recorded in `sec_user_scope` with `GrantedBy` and `GrantReason`, and recertified quarterly by the relevant department head with a fourteen-day response window; scopes not recertified expire automatically.

3.8.2 Audit logging

HIPAA requires that the network be able to reconstruct who accessed PHI, when, and what they did. Four log sources are combined and exported daily to the enterprise SIEM, where they are retained for six years to satisfy the HIPAA documentation retention requirement.

Source	Captures	Retention	Reviewed By
Microsoft Purview audit log (Fabric activity)	Report views, semantic model queries, item creation, sharing, permission changes, exports	6 years in SIEM	Privacy Officer, monthly

Source	Captures	Retention	Reviewed By
Entra ID sign-in and PIM activation logs	Authentication, MFA, conditional access outcome, privileged role activation with justification	6 years in SIEM	Security Officer, monthly
SQL endpoint query log	Direct T-SQL queries against gold, including the text of queries touching the vault schema	6 years in SIEM	Privacy Officer, weekly for vault access
Pipeline run log (audit schema)	Row counts in and out, quarantine volumes, failures, and the identity that executed the run	3 years	Data Engineering, daily

Two detective controls run on top of these logs. Any query against `dim_patient_vault` raises an alert to the Privacy Officer within fifteen minutes, because legitimate access to that table is rare and should be individually explainable. And a weekly anomaly report flags users whose volume of distinct patients viewed exceeds three standard deviations above their own trailing 90-day baseline, which is the pattern that identifies both curiosity browsing and compromised credentials.

3.8.3 Breach notification procedure

The platform does not create a separate breach process; it feeds the network existing HIPAA incident process, and this section defines the platform obligations within it.

1. Detection and containment (target: within 1 hour of detection). The on-call data engineer suspends the affected access — removing the group assignment, disabling the security policy, or pausing the workspace — and preserves the relevant logs by exporting the affected time window to immutable storage before any remediation changes the state.
2. Notification to the Privacy Officer (within 1 hour). Any suspected impermissible access or disclosure of PHI is reported immediately, without waiting for confirmation. Under-reporting a false alarm is cheap; late reporting is not.
3. Risk assessment (within 5 business days). The Privacy Officer conducts the four-factor assessment required by the Breach Notification Rule: nature and extent of the PHI involved, the unauthorized person who accessed it, whether PHI was actually acquired or viewed, and the extent to which risk has been mitigated. The platform supplies the evidence: the exact rows and columns returned by the queries in question, reconstructed from the query log and the state of the RLS scope table at that time.
4. Notification (as determined by assessment). If a breach is confirmed, affected individuals are notified without unreasonable delay and no later than sixty days from discovery; the Secretary of Health and Human Services is notified within sixty days for breaches affecting five hundred or more individuals, and through the annual log otherwise; media notification applies to breaches affecting more than five hundred residents of a state or jurisdiction.
5. Post-incident review (within 30 days). A root-cause review determines whether a control failed or was absent, and produces a tracked remediation item. Where the cause was a permission misconfiguration, the reconciliation audit in Section 3.7.2 is extended to detect that class of error.

3.8.4 Data retention and disposal

Data	Retention	Disposal Method
Bronze raw clinical extracts	7 years, cool tier after 13 months	Automated Delta VACUUM and partition drop; deletion certificate logged
Silver conformed clinical data	7 years	Partition drop on schedule; verified against retention register
Gold facts and dimensions	7 years rolling	Partition drop; aggregate history retained indefinitely as it is de-identified
PHI vault schema	7 years, reviewed annually	Partition drop with Privacy Officer sign-off
Eventhouse real-time events	90 days	KQL retention policy, automatic
Audit and access logs	6 years (HIPAA documentation requirement)	Immutable SIEM storage; no early deletion permitted
Quarantine and rejected records	2 years	Automatic purge after steward disposition
Development and test data	De-identified or synthetic only; refreshed quarterly	Workspace-level purge on refresh

Note the constraint in the last row. Production PHI is never copied to development or test. Test uses a masked, referentially consistent dataset generated by an approved de-identification job, which preserves data volume and distribution for performance validation without carrying identifiers into an environment where eight engineers hold Contributor.

3.8.5 Additional technical safeguards

- Encryption: data at rest in OneLake is encrypted by the platform; the network additionally applies customer-managed keys where supported. Data in transit uses TLS 1.2 or higher on every connection, including the EHR extract.
- Network isolation: capacity access is restricted to the corporate network and managed devices through conditional access policy; a private endpoint is used for the EHR source connection so extract traffic does not traverse the public internet.
- Export controls: the tenant-level setting permitting export to Excel and CSV is disabled for the clinical workspaces, and Purview sensitivity labels with encryption are applied to the semantic model so that any file that is legitimately exported carries protection with it.
- Sharing controls: publish-to-web is disabled tenant-wide; external guest sharing is blocked for the clinical workspaces; and reports may not be shared to distribution lists that are not backed by a recertified security group.

3.9 Data Governance

Governance roles are assigned by name before go-live. The Chief Medical Information Officer owns KPI definitions and adjudicates disputes about clinical meaning. A named data steward in clinical informatics owns terminology mappings and the quarantine queue. The Privacy Officer owns the PHI classification in

Section 3.4 and approves any change to it. Data Engineering owns pipeline correctness and the operational log.

The semantic model is published with the Certified endorsement so that it appears as the network-approved source in the Fabric data hub; uncertified models built by analysts in the sandbox workspace remain visibly distinct. Item lineage and impact analysis are reviewed before any breaking change to a gold table, and the change advisory process in Section 5.8 applies to any change that alters a published KPI definition.

4. Capacity and Performance

4.1 Workload Profile

Capacity sizing begins with a quantified workload rather than a rule of thumb. The figures below are the planning basis; they should be re-measured after ninety days in production and the SKU revisited against actual consumption.

Dimension	Planning Estimate	Basis
Hospitals	12	Network scope
Encounters per day (all classes)	~9,000	3.3M annual encounters across inpatient, observation, ED, and outpatient
Related detail rows per day	~185,000	Diagnoses, procedures, ADT events, and orders at roughly 20 rows per encounter
Daily incremental volume (compressed)	1.0 – 1.5 GB	Delta Parquet after V-Order, including 48-hour lookback re-processing
Bronze at 7-year retention	~2.6 TB	Daily volume plus historical backload
Silver + gold steady state	~900 GB	Conformed and aggregated layers
Named report consumers	~830	Executives, department heads, providers, analysts
Peak concurrent viewers	150 – 200	Monday 07:00–09:00 across all sites
Typical concurrent viewers	25 – 45	Weekday business hours outside peak
Scheduled batch window	01:00 – 05:30	Nightly load, must complete before 06:00 freshness SLA
Real-time event rate	~52,000 ADT messages/day, peak ~120/min	Existing integration engine throughput

4.2 SKU Recommendation

Recommendation: F64 for production, F16 for test, F8 for development, with the production capacity purchased as a one-year reservation and burst headroom handled as described in Section 4.4.

SKU	Capacity Units	Assessment for This Workload	Decision
F32	32 CU	Sufficient for the nightly batch in isolation, but leaves little margin for the Monday morning interactive peak alongside the streaming workload. Critically, it sits below the free-viewer threshold, so every one of roughly 830 consumers would require an individual Power BI Pro license.	Rejected
F64	64 CU	Comfortably absorbs the batch window, the 150–200 concurrent-viewer peak, and continuous Eventstream ingestion with headroom for growth. Meets the threshold at which report consumers do not require individual Pro licenses, which is decisive at this user count. Supports Direct Lake at the gold table sizes modelled here.	Recommended

SKU	Capacity Units	Assessment for This Workload	Decision
F128	128 CU	Over-provisioned against the measured profile; roughly double the cost for headroom that scheduled scaling can supply for six hours a week instead of continuously.	Rejected as a base; used as a scheduled burst target

4.2.1 Rationale

1. Licensing economics dominate. Below the F64 threshold, every consumer needs a Power BI Pro license. At roughly 830 named consumers, the annual per-user cost exceeds the difference between F32 and F64 several times over. The correct comparison is not F32 capacity cost against F64 capacity cost, but F32 plus 830 Pro licenses against F64 alone — and F64 wins decisively.
2. The peak, not the average, sizes the capacity. Average daily utilization is projected at roughly fifteen to twenty percent of an F64. That number is not the constraint. Monday between 07:00 and 09:00, when every department reviews weekend activity simultaneously, is the constraint, and F32 would risk throttling exactly when the platform is most visible to clinical leadership.
3. Direct Lake requires memory headroom. Direct Lake loads column segments into capacity memory on demand. The gold model at roughly 900 GB across all layers, with an active working set well under that, fits F64 memory limits with margin; F32 would risk fallback to DirectQuery on wide queries, which would silently degrade the response times committed in Section 4.5.
4. The streaming workload runs continuously. Eventstream and Eventhouse consume capacity around the clock, unlike the batch and interactive workloads. That baseline consumption must be subtracted before assessing peak headroom, and it is what pushes a marginal F32 case over the line.
5. Non-production environments do not need production capacity. Development at F8 and test at F16 are sized for functional correctness and, in test, for a proportional performance signal. Both are paused outside working hours, which is discussed under cost management below.

4.2.2 Indicative capacity unit budget (production, typical weekday)

Workload	Window	Estimated CU-seconds	Share of F64 daily budget
Spark notebooks (silver + gold build)	01:00 – 04:30	~115,000	2.1%
Copy Job and pipeline activities	01:00 – 02:00	~25,000	0.5%
Semantic model framing and warm-up	05:00 – 05:30	~15,000	0.3%
Interactive queries — peak two hours	07:00 – 09:00	~260,000	4.7%
Interactive queries — remaining day	09:00 – 19:00	~180,000	3.3%
Eventstream + Eventhouse (continuous)	24 hours	~260,000	4.7%

Workload	Window	Estimated CU-seconds	Share of F64 daily budget
Real-time dashboard queries	24 hours	~45,000	0.8%
Total		~900,000 of 5,529,600	~16%

The sixteen percent figure is a daily average and should not be read as spare capacity available for other workloads. Interactive demand is concentrated: the 260,000 CU-seconds in the peak two hours represent roughly thirty-six percent utilization during that window, and Monday runs materially higher than a typical weekday. The budget is presented so that the review board can see both the comfortable average and the reason the peak still governs the decision.

4.3 Cost Management

- Reserve the production F64 for one year, which reduces the rate materially against pay-as-you-go for a workload that runs continuously.
- Run development and test as pay-as-you-go and pause them automatically outside 07:00–19:00 on weekdays. A paused capacity accrues no compute charge, and the pause and resume are scripted through the Azure Resource Manager API as part of the same automation used for scheduled scaling.
- Budget OneLake storage separately from capacity. Storage is billed on volume, not on SKU, so retention policy is a cost lever independent of capacity sizing — moving bronze to cool tier after thirteen months is a meaningful saving at 2.6 TB.
- Charge back to hospital sites on report consumption metrics rather than splitting the capacity cost evenly, so that sites see the relationship between usage and cost.

4.4 Autoscaling Strategy

Fabric capacity does not scale continuously on demand in the way an application tier does. The strategy therefore combines four mechanisms — two built into the platform, two implemented by the network — to absorb the Monday peak without paying for F128 for one hundred and sixty-eight hours a week.

4.4.1 Mechanism 1 — Platform smoothing and bursting (built in)

Fabric smooths background operations across a twenty-four-hour window and interactive operations across a five-minute window, and allows short bursts above the purchased capacity level to be repaid from the smoothed budget. In practice this means the nightly Spark load does not need to fit inside 64 CU at any single instant; it needs to fit inside the daily budget. No configuration is required, but the behaviour must be understood, because it is why the batch window in Section 4.2.2 consumes so little of the headroom that matters.

4.4.2 Mechanism 2 — Scheduled vertical scaling (implemented)

An Azure Automation runbook, triggered on a schedule, calls the Fabric capacity management API to change the SKU. The scale operation is online and takes effect within a few minutes.

Schedule	Action	Reason
Monday 05:30 → F128	Scale up	Weekend review peak; the heaviest interactive load of the week
Monday 11:00 → F64	Scale down	Peak has passed; demand returns to typical weekday levels
Month-end close, 07:00 → F128	Scale up	Board and service-line reporting cycle, three business days
Month-end close +3 days → F64	Scale down	Cycle complete
Dev / Test 19:00 → paused	Pause	No development activity overnight or at weekends
Dev / Test 07:00 weekdays → resumed	Resume	Working hours

Scale-down is never automatic while a long-running job is in flight; the runbook checks for active Spark sessions and pipeline runs and defers by fifteen minutes, up to three attempts, before alerting rather than interrupting a load.

4.4.3 Mechanism 3 — Reactive scaling on sustained pressure (implemented)

The scheduled pattern handles known peaks. Unknown peaks — an unplanned surge, a poorly written report published by an analyst, a mass-casualty event driving unusual dashboard traffic — are handled reactively. An alert on sustained capacity utilization above eighty-five percent for fifteen minutes triggers the same runbook to scale to the next SKU and notify the on-call engineer, who then investigates the cause. Automatic scale-down follows two hours after utilization returns below sixty percent, subject to the same in-flight-job check.

4.4.4 Mechanism 4 — Surge protection and workload isolation (configured)

Surge protection is enabled on the production capacity so that background operations are rejected once background utilization crosses a defined threshold, protecting interactive users from being throttled by a runaway batch job. In addition, the sandbox workspace where analysts explore de-identified data is assigned to the development capacity rather than production, so exploratory work cannot consume capacity that clinical users depend on. This is a deliberate architectural choice: the cheapest form of autoscaling is preventing the load in the first place.

4.5 Performance Targets

These targets are service-level objectives, measured monthly and reported to the Director of Enterprise Data and Analytics. Breaching a target for two consecutive months triggers a formal performance review, not merely a ticket.

Metric	Target	Alert Threshold	Measured By
Report page load, initial render	p95 ≤ 5 seconds	> 8 seconds sustained 15 min	Capacity Metrics app, interactive operations
Visual interaction (slicer, cross-filter)	p95 ≤ 3 seconds	> 5 seconds sustained 15 min	Capacity Metrics app
Direct Lake queries falling back to DirectQuery	0%	Any occurrence	Semantic model query log
Nightly refresh duration (end to end)	≤ 3.5 hours	> 4 hours	Pipeline run log
Data freshness — gold available	By 06:00 local, 7 days a week	Not complete by 06:00	Activator freshness rule
Real-time census latency (event to dashboard)	≤ 5 minutes	> 10 minutes	Eventhouse ingestion latency metric
Interactive capacity utilization	≤ 70% sustained	> 85% for 15 min → scale	Capacity Metrics app
Background capacity utilization	≤ 60% sustained	> 80% for 30 min	Capacity Metrics app
Throttling / delayed operations	0 minutes per month	Any occurrence	Capacity Metrics app, throttling page
Platform availability (business hours)	99.9%	Two consecutive failed health checks	Synthetic report query every 5 min
Silver quarantine rate	≤ 0.5% of daily rows	> 1% of daily rows	Pipeline run log

4.5.1 Engineering practices supporting these targets

- V-Order and OPTIMIZE with compaction run after every gold load, so Direct Lake reads a small number of well-sized row groups. Small-file proliferation is the most common cause of Direct Lake performance decay.
- Gold facts are partitioned by discharge date and census date; date-filtered queries — which is nearly all of them — prune at the file level.
- The agg_kpi_dept_day table backs the highest-traffic report pages so that the executive summary page never scans the encounter fact.
- Report design standards limit each page to eight visuals, prohibit table visuals over ten thousand rows, and require that default slicer state restrict to the trailing thirteen months.
- Every report is profiled with Performance Analyzer before promotion to production, with results attached to the pull request; a page exceeding the p95 target is not promoted.

5. DevOps and Deployment

5.1 Environment Model

Three environments, each a separate Fabric workspace on a separate capacity, with a strictly one-directional promotion path. Nothing reaches production except through this path.

Environment	Workspace	Data	Who Has Write Access	Purpose
Development	ws-clinical-dev	Synthetic + de-identified sample	Engineers (Contributor)	Build and unit test
Test	ws-clinical-test	Masked, production-scale volume	Deployment service principal only	Integration, performance, security validation, stakeholder acceptance
Production	ws-clinical-prod	Live PHI	Deployment service principal only	Clinical use

5.2 Git Branching Strategy

A feature-branch workflow on a single long-lived branch. The repository is hosted in Azure DevOps and connected to Fabric through native Git integration.

```

main                protected; always deployable; mirrors production
|
+-- feature/DA-142-alos-by-service-line      new work, short-lived
+-- feature/DA-155-readmission-fact
+-- bugfix/DA-161-null-discharge-disposition
+-- hotfix/DA-170-rls-scope-expiry           branched from main,
|                                             merged back immediately
+-- release/2026.09                         optional, cut only when a
                                             release must be stabilized
                                             while main moves ahead

```

5.2.1 Rules

- Branch naming is enforced: feature/, bugfix/, hotfix/, or release/ followed by the work item identifier and a short description. The work item link is what ties a production change back to a clinical requirement during audit.
- Feature branches are short-lived — three working days is the target, five the limit — because long-lived branches on a shared semantic model produce merge conflicts in model metadata that are painful to resolve.
- Each engineer works in a personal Fabric workspace connected to their own feature branch, so that in-progress work never destabilizes the shared development workspace.
- main is protected: no direct commits, no force push, linear history required.

5.2.2 Pull request policy

Policy	Setting	Reason
Minimum reviewers	2	One peer engineer plus one domain reviewer
Required reviewer — semantic model changes	Analytics engineering lead	Protects certified KPI definitions from drift
Required reviewer — anything touching gold.dim_patient_vault, sec_user_scope, or the sec schema	Privacy Officer delegate	A change to PHI handling or entitlement logic is a compliance change, not just a code change
Linked work item	Required	Traceability from production behaviour to approved requirement
Build validation	Required, must pass	Static analysis and model best-practice rules
Comment resolution	All comments resolved	Prevents merge with open objections
Self-approval	Prohibited	Separation of duties

5.3 Fabric Git Integration

The development workspace is connected to main. Engineers sync their personal workspaces to their feature branches, commit item definitions — notebooks as .py or .ipynb, pipelines as JSON, semantic models as TMDL, reports as .pbir project folders — and open a pull request. The .pbir and TMDL formats matter here: they store report and model definitions as readable, diffable text rather than as opaque binary .pbix files, which is what makes meaningful code review of a semantic model change possible at all.

Lakehouse table data is not source-controlled. Only the definitions — schemas, notebooks, pipeline logic, security policy scripts — are versioned. The T-SQL security scripts in Section 3.7 are held in the repository under /sql/security/ and applied by the release pipeline, so that permissions are code and are reviewed as code.

5.4 Deployment Pipeline Design

A Fabric deployment pipeline with three stages moves content between workspaces. Item identifiers are preserved across stages, so a report deployed to test retains its binding to the test semantic model rather than pointing back to development.

Stage	Source	Trigger	Approval	Gate Before Promotion
Development	main branch sync	Automatic on merge to main	None	Build validation passed
Test	Dev stage	Automatic after Dev deploy succeeds	None (automated)	Automated test suite: data quality, RLS/OLS assertions, performance profile
Production	Test stage	Manual release, scheduled window	Analytics lead + Privacy Officer for PHI-affecting changes	Stakeholder acceptance sign-off, change record approved

Production deployments occur in a defined window — Tuesday and Thursday, 20:00 to 22:00 local — outside the nightly load and away from the Monday peak. Emergency changes bypass the window only under the hotfix path in Section 5.8.

5.5 Deployment Rules for Environment Differences

The same artifact is promoted unchanged through all three stages; only the bindings differ. Those differences are expressed as deployment rules on the pipeline and as pipeline parameters, never as edited copies of items — an edited copy in production is an untracked divergence and defeats the entire promotion model.

Item	Setting	Development	Test	Production
cj_ehr_encounters	EHR connection	ehr-dev.internal (synthetic)	ehr-tst.internal (masked)	ehr-prd.internal (live)
pl_ehr_daily_load	Parameter: env	dev	tst	prd
pl_ehr_daily_load	Parameter: lookback_hours	24	48	48
pl_ehr_daily_load	Parameter: hospital_list	2 sites	All 12	All 12
pl_ehr_daily_load	Schedule	Manual only	Nightly 22:00	Nightly 01:00
sm_clinical_ops	Lakehouse binding	lh_gold (dev)	lh_gold (test)	lh_gold (prod)
sm_clinical_ops	RLS role membership	Test principals	Test principals	Live Entra groups
rpt_* (all reports)	Semantic model binding	Dev model	Test model	Prod model
es_adt_stream	Source endpoint	Message simulator	Integration engine (test queue)	Integration engine (live)
act_ops_alerts	Notification target	Engineer inbox	Engineer inbox	House supervisor Teams channel + on-call
All items	Sensitivity label	Internal	Internal	Confidential — PHI

The alert-routing rule in the second-to-last row is easy to overlook and expensive to get wrong: without it, a test deployment pages the house supervisors at every hospital in the network.

5.6 CI/CD Trigger Configuration

Two Azure DevOps pipelines. A build pipeline runs on every pull request against main and blocks the merge on failure. A release pipeline runs on completion of that pull request — that is, on merge — and drives the Fabric deployment pipeline through the REST API using the deployment service principal.

```
# azure-pipelines-build.yml (pull request validation)
pr:
  branches:
    include: [ main ]
```

```

stages:
- stage: Validate
  jobs:
  - job: StaticChecks
    steps:
    - script: |
      # Semantic model: best practice rules and definition validation
      TabularEditor.exe ./model/sm_clinical_ops.tmdl \
      -A ./rules/BPARules-clinical.json -V
      displayName: Semantic model best practice analyzer

    - script: |
      # Report definitions: .pbir schema validation and theme compliance
      python ./ci/validate_pbir.py ./reports/
      displayName: Validate .pbir report definitions

    - script: |
      # Notebooks: lint and unit tests for transformation logic
      pytest ./tests/unit -q
      displayName: Notebook unit tests

    - script: |
      # Security: fail the build if a PHI column appears outside the
      # vault schema or an RLS role is missing a filter expression
      python ./ci/check_phi_exposure.py --model ./model/ \
      --classification ./governance/phi_columns.yml
      displayName: PHI exposure check

# azure-pipelines-release.yml (runs on PR completion / merge to main)
trigger:
  branches:
    include: [ main ]

stages:
- stage: DeployDev
  jobs:
  - job: SyncAndDeploy
    steps:
    - script: |
      # Update the dev workspace from main, then promote Dev -> Test
      fab api workspaces/${devWorkspaceId}/git/updateFromGit --method post
      fab api deploymentPipelines/${pipelineId}/deploy --method post \
      --body @deploy-dev-to-test.json
      displayName: Sync dev workspace and deploy to Test

- stage: TestGates
  dependsOn: DeployDev
  jobs:
  - job: AutomatedValidation
    steps:
    - script: pytest ./tests/data_quality -q
      displayName: Data quality assertions against test gold layer
    - script: python ./ci/test_qls_roles.py --workspace ${testWorkspaceId}
      displayName: RLS and OLS assertions for every role
    - script: python ./ci/test_performance.py --p95-seconds 5
      displayName: Report performance profile

- stage: DeployProd
  dependsOn: TestGates
  condition: succeeded()
  jobs:
  - deployment: PromoteToProduction
    environment: fabric-clinical-prod # carries the manual approval gate
    strategy:
      runOnce:
        deploy:
          steps:
          - script: |
              fab api deploymentPipelines/${pipelineId}/deploy --method post \
              --body @deploy-test-to-prod.json
              displayName: Promote Test to Production
          - script: |
              # Apply versioned SQL security objects to the prod endpoint
              sqlcmd -S ${prodSqlEndpoint} -G -i ./sql/security/*.sql
              displayName: Apply GRANT/DENY and security policies
          - script: python ./ci/smoke_test.py --workspace ${prodWorkspaceId}
              displayName: Post-deployment smoke test

```

Semantic model deployment uses the XMLA endpoint for anything the deployment pipeline cannot express — partition definitions, OLS role metadata, and incremental refresh policies are applied by a Tabular Editor script over XMLA rather than by hand. Report deployment uses the .pbir project format, so reports are deployed as definitions bound to the target model rather than as republished .pbix files carrying an embedded dataset.

5.7 Release Gates

A build reaching production must have passed all of the following. These are automated except where noted.

Gate	Stage	Failure Behaviour
Model best-practice analyzer, zero errors	PR validation	Blocks merge
PHI exposure check — no classified column outside the vault schema	PR validation	Blocks merge
Notebook unit tests	PR validation	Blocks merge
Row-count and referential integrity assertions on test gold	Test	Blocks promotion
RLS assertion suite: each role returns exactly the expected row count for a set of test principals, including a principal with no scope returning zero rows	Test	Blocks promotion
OLS assertion suite: restricted columns are absent from metadata for each role	Test	Blocks promotion
Report performance profile within p95 target	Test	Blocks promotion
Stakeholder acceptance sign-off (manual)	Test	Blocks promotion
Change record approved by change advisory board (manual)	Production	Blocks deployment
Post-deployment smoke test: model queryable, reports render, freshness rule green	Production	Triggers rollback

The RLS assertion suite deserves particular emphasis. Row-level security is the control most likely to be silently broken by an unrelated model change — adding a relationship, changing a filter direction, renaming a dimension — and the failure is invisible in normal use because the user who is over-permissioned sees more data without any error. Automated assertion is the only reliable defence.

5.8 Change Management in a Clinical Environment

- Standard changes — new report page, new measure that does not alter an existing definition, cosmetic changes — follow the normal pipeline with the deployment window.

- Significant changes — any change to a published KPI definition, to the security model, or to gold table schema — require change advisory board approval and advance notice to affected clinical leadership, because a leader who sees their length-of-stay figure move overnight without explanation will lose trust in the platform regardless of whether the new figure is more correct.
- Emergency changes — a defect producing clinically misleading output, or an access control failure — may bypass the window under the hotfix path, with retrospective approval within one business day and a mandatory post-incident review. An access control failure is contained first and fixed second: revoke access, then diagnose.
- Rollback is by redeploying the previous tagged release through the same pipeline. Every production deployment is tagged in Git with the release identifier and the change record number, so rollback is a documented, tested path rather than an improvisation at 22:30.

6. Operational Procedures

6.1 Capacity Monitoring

The Microsoft Fabric Capacity Metrics app is the primary instrument, supplemented by pipeline run logs and Activator rules. Monitoring is deliberately narrow — a small number of metrics with defined thresholds and a named recipient — because an alert that nobody owns is noise, and noise in a 24/7 environment trains people to ignore pages.

Metric	Warning	Alert / Action	Notified
Interactive CU utilization	> 70% for 30 min	> 85% for 15 min → auto-scale + page	On-call engineer
Background CU utilization	> 60% for 60 min	> 80% for 30 min → investigate	Data engineering channel
Throttling / delayed operations	Any	Any occurrence → Sev 2 incident	On-call + analytics lead
Overage / carry-forward debt	> 10% of daily budget	> 25% → scale review within 24h	Analytics lead
Nightly pipeline duration	> 3 hours	> 4 hours → investigate before 06:00	On-call engineer
Pipeline failure	—	Any failure → Sev 1 or 2 per Section 6.2	On-call engineer
Data freshness (gold updated)	—	Not complete by 06:00 → Sev 1	On-call + clinical ops duty manager
Silver quarantine rate	> 0.5% of rows	> 1% → data steward review same day	Clinical informatics steward
Direct Lake fallback events	—	Any occurrence → model review	Analytics engineering
Real-time ingestion lag	> 5 min	> 10 min → Sev 2	On-call + integration team
PHI vault query executed	—	Any occurrence → review within 15 min	Privacy Officer
Failed authentication spike	> 10 in 5 min for one principal	Conditional access lockdown + review	Security operations

A weekly capacity report is generated from the Metrics app and reviewed at the Monday operations stand-up: peak utilization by day, top ten CU-consuming items, slowest report pages, and trend against the ninety-day baseline. The purpose of this review is to catch gradual degradation — a report that grew from three seconds to seven over two months — before it becomes an incident.

6.2 Incident Response

6.2.1 Severity definitions

Severity	Definition	Response Target	Escalation
Sev 1	PHI exposure or suspected impermissible access; clinically misleading data published; platform unavailable during business hours; data not refreshed by 06:00	15 min acknowledge, 1 hour containment	Immediate to analytics lead; Privacy Officer if PHI involved
Sev 2	Pipeline failure with refresh recoverable before 06:00; sustained throttling; real-time path down; single report broken	30 min acknowledge, 4 hour resolution	Analytics lead at 4 hours
Sev 3	Partial data quality issue affecting one department; slow but functional reports; non-urgent defect	Next business day	Standard backlog

6.2.2 Runbook — nightly pipeline fails at 03:00

This is the scenario the on-call rotation exists for, and it is worth walking through concretely because the decisions are time-boxed against the 06:00 clinical deadline.

1. 03:00 — Detection. The pipeline On Failure activity writes to the operational log, posts to the data engineering Teams channel, and raises a PagerDuty incident. Activator independently monitors freshness, so a pipeline that hangs without failing is still detected.
2. 03:05 — Acknowledgement. The on-call data engineer acknowledges within the fifteen-minute target. Failure to acknowledge escalates automatically to the secondary on-call at fifteen minutes and to the analytics lead at thirty.
3. 03:05–03:20 — Triage. The engineer identifies the failed activity and classifies the cause into one of four buckets: source unavailable (EHR maintenance window overran, credential expired, network path down); data quality (schema drift, unmapped codes above threshold, referential failure); capacity (throttling, Spark session exhaustion); or platform (Fabric service incident, checked against the service health dashboard).
4. 03:20 — Decision point. If the failure is transient, the engineer reruns from the failed activity; the pipeline is idempotent and restartable at activity granularity, which is why it is built that way. If the rerun can complete before 05:30, the 06:00 SLA is still met and the incident closes as Sev 2. If it cannot, the incident is raised to Sev 1 and step 5 applies.
5. 04:00 — Partial publication. Where only some hospitals failed, the pipeline publishes the successful sites and marks the failed sites as stale in a data-status table that every report surfaces as a banner. Publishing eleven of twelve hospitals with a visible warning is better than publishing nothing, and far better than publishing twelve hospitals of which one is silently wrong.
6. 05:30 — Communication. If the 06:00 SLA will be missed, the on-call engineer notifies the clinical operations duty manager and posts to the clinical leadership channel with a plain statement of what is stale, which sites are affected, and the expected recovery time. Communication is not optional and is not delegated to the recovery being finished.
7. 08:00 — Handover. Unresolved incidents transfer to the day team at the morning stand-up with a written handover: what happened, what has been tried, current state, next action.

8. Within 5 business days — Post-incident review. Sev 1 and repeated Sev 2 incidents receive a blameless review producing a tracked action. Recurring causes drive automation: three occurrences of the same failure mode is treated as a defect in the pipeline, not in the on-call rotation.

6.2.3 On-call model for a 24/7 environment

Tier	Who	Coverage	Responsibility
Primary	Data engineering rotation (8 engineers, weekly)	24/7	First response, triage, rerun, containment
Secondary	Senior data engineer	24/7 backup	Escalation at 15 min; complex recovery
Analytics lead	Named individual	Business hours + Sev 1 any time	Decision authority on partial publication and clinical communication
Privacy Officer	Named individual	Sev 1 PHI incidents, any time	Breach assessment and notification decision
EHR application on-call	EHR team rotation	24/7	Source-side failures
Microsoft support	Pro Direct / Premier	24/7 for Sev A	Platform incidents

The on-call engineer is compensated for the rotation and is not expected to work a normal day after a night incident. This is stated in an architecture document deliberately: a 24/7 support commitment that depends on unpaid goodwill degrades within a quarter, and the operational SLAs above are only credible if the rotation is genuinely staffed.

6.3 Scheduled Maintenance

Activity	Frequency	Owner	Output
Capacity utilization and performance review	Weekly (Mon)	Analytics lead	Trend report; scaling recommendation
Delta OPTIMIZE and VACUUM on gold	Weekly (Sun 02:00)	Automated	Compaction log
Quarantine queue disposition	Daily	Clinical informatics steward	Cleared queue; mapping updates
Backup and restore test (workspace + model)	Monthly	Data engineering	Verified restore evidence
SQL / workspace permission reconciliation	Monthly	Security Officer	Discrepancy report; 14-day remediation
Access recertification (sec_user_scope)	Quarterly	Department heads	Confirmed or revoked scopes; expiry applied
SKU and cost review against actuals	Quarterly	Analytics lead + Finance	Reservation and scaling decision

Activity	Frequency	Owner	Output
Security audit — RLS/OLS/CLS effectiveness testing	Semi-annual	Security + Privacy Officers	Audit report to compliance committee
HIPAA risk assessment refresh	Annual	Privacy Officer	Updated risk register
Disaster recovery exercise	Annual	Data engineering + Security	Tested runbook; RTO/RPO evidence
Architecture document review	Semi-annual, or on major change	Data platform architect	Revised version through document control
KPI definition review with CMIO	Quarterly	CMIO + analytics lead	Confirmed or revised definitions

6.4 Business Continuity

The platform is an operational reporting system, not a clinical system of record. Its unavailability does not affect patient care directly, and the recovery objectives reflect that proportionality: a recovery time objective of four hours during business hours and a recovery point objective of one business day, since the authoritative clinical data remains in the EHR and can be re-extracted.

- Item definitions are recoverable from Git; a workspace can be rebuilt from the repository and the deployment pipeline in under two hours, which is verified by the annual exercise.
- Gold data is recoverable by re-running the transformation notebooks from bronze; bronze is recoverable by re-extracting from the EHR within the source retention window.
- During an extended outage, hospitals fall back to EHR-native operational reports. This fallback is documented in the department runbooks so that it is a known procedure rather than an improvisation.

Appendix A — Glossary

Term	Definition
ADT	Admit, Discharge, Transfer — the HL7 message family describing patient movement
ALOS	Average length of stay, computed as patient days divided by discharges
CLS	Column-level security — restricting access to specific columns
CU	Capacity Unit — the unit of Fabric compute; an F64 provides 64
Direct Lake	Fabric semantic model mode reading Delta tables from OneLake without import or DirectQuery
EHR	Electronic Health Record system
EMPI	Enterprise Master Patient Index — resolves one patient identity across sites
HIPAA	Health Insurance Portability and Accountability Act of 1996
LWBS	Left without being seen — an ED throughput measure
Medallion	Bronze / silver / gold layered data architecture
OLS	Object-level security — restricting the visibility of tables or columns in a semantic model
OneLake	The single logical data lake underlying all Fabric workspaces
PHI	Protected Health Information as defined by HIPAA
RLS	Row-level security — restricting which rows a user can retrieve
SCD-2	Slowly changing dimension type 2 — retains history by versioning rows
SIEM	Security Information and Event Management system
TMDL	Tabular Model Definition Language — text format for semantic model definitions
V-Order	Fabric write-time optimization improving Delta Parquet read performance
XMLA	The endpoint protocol used for programmatic semantic model deployment
.pbir	Power BI report project format storing report definitions as text for source control

Appendix B — KPI Definitions

These definitions are the network-approved standard and are implemented once, in the certified semantic model. Any report displaying a differently-defined measure is out of compliance with this architecture.

KPI	Numerator	Denominator	Inclusions / Exclusions
Average length of stay	Sum of patient days for discharged encounters	Count of discharges	Inpatient and observation only; excludes same-day discharges and encounters with LOS > 365 days pending review

KPI	Numerator	Denominator	Inclusions / Exclusions
30-day unplanned readmission rate	Count of index discharges followed by an unplanned inpatient readmission within 30 days	Count of eligible index discharges	Excludes planned readmissions, transfers to another acute facility, deaths, and discharges against medical advice
Encounter volume by department	Count of encounters	n/a	Attributed to the discharging department; transfers counted once
Occupancy rate	Sum of midnight census	Sum of staffed beds	Staffed beds sourced from the daily capacity file, not licensed beds
ED length of stay	Sum of minutes from arrival to departure	Count of ED encounters	Separated into discharged and admitted cohorts; boarding time reported separately
Observation conversion rate	Count of observation encounters converted to inpatient	Count of observation encounters	Conversion identified by encounter class change within the same episode

Appendix C — Open Items for Review Board Decision

#	Question	Owner	Needed By
O1	Confirm the EHR extract method and whether change-data-capture is available, which determines the achievability of the 3.5-hour refresh target	EHR Application Manager	Before build
O2	Confirm whether providers may see medical record numbers for their own panel, as proposed in Section 3.6.1	Privacy Officer + CMIO	Before build
O3	Agree the small-cell suppression threshold; 11 is proposed, some networks use 6 or 20	Privacy Officer	Before build
O4	Confirm the source of staffed-bed counts for the occupancy denominator	Hospital Operations	Before gold build
O5	Approve the F64 reservation and the F128 scheduled burst budget	Finance + Director EDA	Before capacity purchase
O6	Confirm 24/7 on-call staffing and compensation model for the data engineering rotation	Director EDA + HR	Before go-live
O7	Determine whether release 1 includes outpatient encounters or restricts to inpatient, observation, and ED	CMIO	Before build

End of Document

This architecture is submitted for Architecture Review Board consideration. Approval of this document authorizes the build described in Sections 2 through 6; it does not authorize the processing of live PHI,

which requires separate confirmation that the items in Appendix C are closed and that the Privacy and Information Security Officers have signed the control attestations in Section 3.